

Linux Server Hardening & Deployment Checklist

Free Resource by Dhantel R. Parwedha

Version: 1.0

Target OS: Ubuntu Server 22.04+, Debian 12+, Rocky Linux 9+

1. Initial Server Setup

Server Information

Item	Status
Hostname configured	☐
Timezone configured	☐
NTP synchronization active	☐
Static IP configured	☐
DNS configured	☐

Commands

```
hostnamectl
timedatectl
ip a
resolvectl status
```

2. User & Access Management

User Security

Check	Status
Non-root admin account created	☐
Strong password applied	☐
Sudo privileges configured	☐
Unused users removed	☐

Commands

```
adduser adminuser
usermod -aG sudo adminuser
```

3. SSH Hardening

SSH Configuration

Check	Status
Root login disabled	☐
Password authentication disabled	☐
SSH keys configured	☐
SSH port customized	☐
Fail2Ban installed	☐

Commented [RP1]: Kenapa penting?

- Memudahkan identifikasi server saat login SSH.
- Muncul di log sistem.
- Memudahkan monitoring dan inventory.
- Sangat membantu jika mengelola banyak server.

Commented [RP2]: Kenapa penting?

Log sistem bergantung pada waktu.

Commented [RP3]: Kenapa penting?

Kalau waktu server meleset:

- SSL/TLS bisa gagal
- Kerberos/Active Directory bisa gagal login
- Database replication bisa bermasalah
- Log antar server tidak sinkron

Commented [RP4]: Kenapa penting?

Server harus bisa ditemukan secara konsisten.

Layanan seperti:

- Web Server
- Database
- DNS
- VPN
- Monitoring

biasanya mengandalkan IP tetap.

Commented [RP5]: Kenapa penting?

Hampir semua layanan modern membutuhkan DNS.

Contoh:

| apt update

akan gagal jika DNS bermasalah.

Begitu juga:

| ping google.com

Kalau DNS rusak:

| Temporary failure in name resolution

Padaahal koneksi internet sebenarnya masih hidup.

Verify

```
cat /etc/ssh/sshd_config
Recommended:
PermitRootLogin no
PasswordAuthentication no
```

4. Firewall Configuration

Network Security

Check	Status
Firewall enabled	☐
Only required ports opened	☐
Unused services blocked	☐

Example

```
ufw allow 22/tcp
ufw allow 80/tcp
ufw allow 443/tcp
ufw enable
```

5. System Updates

Patch Management

Check	Status
Latest updates installed	☐
Security patches applied	☐
Automatic updates configured	☐

Commands

```
Ubuntu:
apt update
apt upgrade -y
Rocky Linux:
dnf update -y
```

6. Monitoring & Logging

Monitoring

Check	Status
CPU monitoring configured	☐
Memory monitoring configured	☐
Disk monitoring configured	☐
Service monitoring configured	☐

Commands

```
top
htop
df -h
journalctl -xe
```

7. Backup Verification

Backup Policy

Check	Status
Backup schedule documented	☐
Backup destination verified	☐
Restore test completed	☐

Verify

```
rsync
restic
borgbackup
```

8. Security Audit

Final Review

Check	Status
No unused services running	☐
Fail2Ban active	☐
Firewall active	☐
Security updates applied	☐
Backups verified	☐
SSH hardened	☐

Quick Audit Commands

```
ss -tulpn
systemctl list-units --type=service
ufw status
fail2ban-client status
```

Monthly Maintenance Checklist

Task	Status
Apply security updates	☐
Review failed login attempts	☐
Verify backup integrity	☐
Check disk usage	☐
Review active services	☐
Review firewall rules	☐
Remove unused accounts	☐

About the Author

Dhantel R. Prawedha is an IT Infrastructure professional with experience in Linux administration, network engineering, server management, virtualization, and hospitality IT operations.

Areas of Expertise

- Linux Server Administration
- Network Infrastructure
- Virtualization & Cloud Services
- System Security & Hardening
- IT Operations Management
- Technical Documentation

Portfolio:

<https://drprawedha.github.io>

GitHub:

<https://github.com/drprawedha>